

Project Workflow & Data Analysis

US Used Cars Dataset (Phases 1–3)



Jyväskylä University of Applied Sciences

Presented by

Group 7

Nirodha Sandaruwani AD9906

Nimesha Fernando - AC5296

Deegoda Gamage - AD9947

Bhagya Collas - AC4885

Nguyen Loi - AC4932

What We Present Today

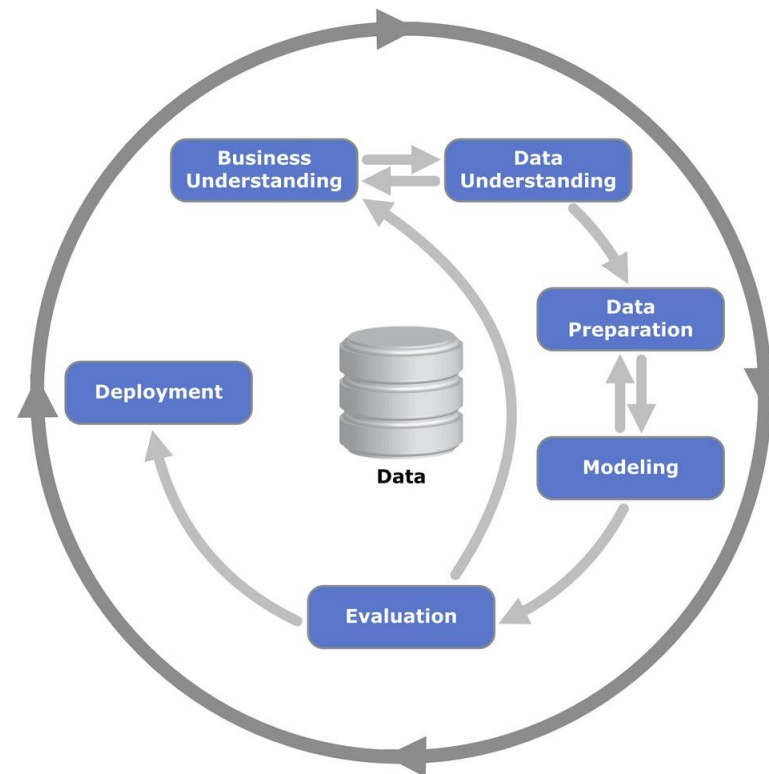
-  **Workflow & Process Management**
-  **Group Work (Iterative)**
-  **Phase 1: Goals & Problem Definition**
-  **Phase 2: Data Understanding**
-  **Phase 3: Data Preparation & Data Analysis**
-  **Notebook Documentation**
-  **Result Analysis**
-  **Next Steps**

Workflow & Process Management

CRISP-DM is a industry-standard approach for data mining projects, providing a structured framework for successful implementation.

Application to US Used Cars Dataset

-  **Business Understanding**
Determining factors influencing used car pricing and identifying market trends
-  **Data Understanding**
Exploring the US Used Cars Dataset features (make, model, year, mileage, condition, etc.)
-  **Data Preparation**
Cleaning missing values, normalizing prices, encoding categorical variables
-  **Modeling**
Implementing regression models to predict prices and classification models for market segmentation
-  **Evaluation**
Assessing model accuracy, RMSE for price predictions, and business value
-  **Deployment**
Creating interactive dashboards and recommendation systems for stakeholders



Group Work & Iterative Progress



Iterative Development

Continuous improvement through feedback loops and testing



Team Collaboration

Using GitHub for version control and code sharing



Weekly Checkpoints

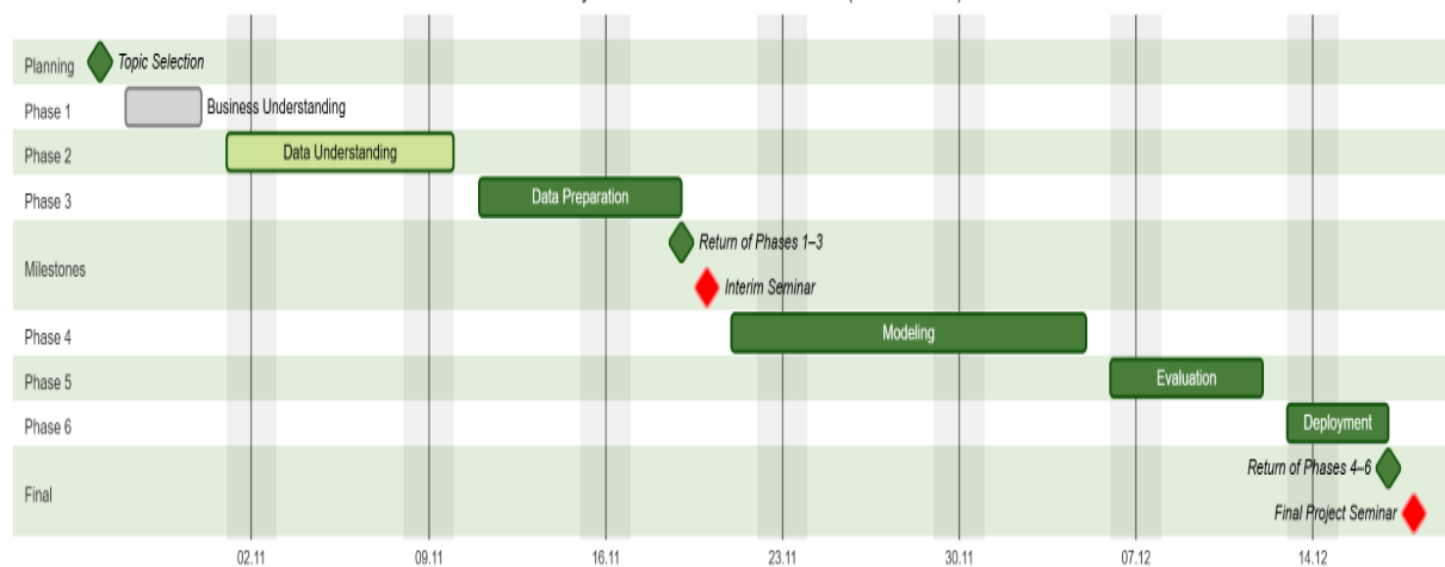
Regular progress reviews and milestone tracking



Task Distribution

Clear assignment of responsibilities and deadlines

Project Timeline - Cars Dataset (CRISP-DM)



Phase 1: Business Understanding

Purpose

Improve pricing strategy, reduce inventory time, and understand selling patterns in the used car market.

Business Objectives



Identify Price Factors

Determine which features most influence used car pricing.



Seller Performance

Compare characteristics of fast vs. slow sellers.



Data-Driven Decisions

Provide analytics to support dealership strategies.



Risks & Constraints

Missing Data

Significant portions of data are missing, requiring careful handling.

Market Evolution

2020 dataset shows changing market behaviors.

Incomplete Fields

Many optional fields are incomplete.



Business understanding drives all subsequent analysis.

Phase 1: Research Questions & Goals

? Primary Research Questions



What factors affect used car prices?

Exploring variables that influence pricing in the used car market



What characteristics determine fast vs slow sellers?

Understanding factors that impact time on market

⚙️ Planned Modeling Tasks & Evaluation

Modeling Approaches



Price prediction (regression)



Sale-time classification (fast/medium/slow)

🎯 Project Analysis Goals

Perform descriptive analysis & visual insights

Understand variable relationships and distributions

Identify predictors for price and days-on-market

Evaluation Metrics

R^2

MAE

RMSE

Accuracy

F1 Score

Dataset Overview



Source

Kaggle — US Used Cars Dataset



Original Dataset

3,000,000+ rows, 66 columns



Working Subset

30,000 rows (1% sample)

For efficient testing and development



Final Pipeline

Designed for full 3M dataset

Scalable processing workflow

shape: (5, 66)

vin	back_legroom	bed	bed_height	...	wheel_system	wheelbase	width	year
---	---	---	---		_display	---	---	---
str	str	str	str		str	str	str	i64
ZACNJABB5KPJ9 2081	35.1 in	null	null	...	Front-Wheel Drive	101.2 in	79.6 in	2019
SALCJ2FX1LH85 8117	38.1 in	null	null	...	All-Wheel Drive	107.9 in	85.6 in	2020
JF1VA2M676982 9723	35.4 in	null	null	...	All-Wheel Drive	104.3 in	78.9 in	2016
SALRR2RV0L243 3391	37.6 in	null	null	...	All-Wheel Drive	115 in	87.4 in	2020
SALCJ2FXXLH86 2327	38.1 in	null	null	...	All-Wheel Drive	107.9 in	85.6 in	2020

Moving to Phase 2 & Phase 3



GitLab Jupyter Notebooks

We are now switching to GitLab Jupyter Notebooks to implement the technical steps for Phase 2 (Data Understanding) and Phase 3 (Data Preparation).

```
# Import libraries and load data
```

```
df = pd.read_csv('used_cars_dataset.csv')
```

```
df.head()
```

Live Demonstration



Data Structure Inspection

Examining schema, data types, and basic statistics



Missing Value Handling

Identifying and addressing missing data patterns



Cleaning & Transformations

Standardizing formats, correcting inconsistencies



Feature Engineering

Creating new meaningful attributes from existing data



Outlier Analysis

Detecting and handling anomalies in the dataset



Final Prepared Datasets

Saving cleaned and engineered datasets for modeling

Summary of Results So Far (Phases 1–3)



Dataset Analysis

Fully explored and analyzed the dataset structure, column types, distributions, and relationships.



Data Quality Issues

Identified missing values, inconsistent units, mixed data formats, duplicated records, and category imbalances.



Data Cleaning

Standardized all key variables by converting units, fixing inconsistent formats, filling missing values, normalizing categories, and ensuring valid boolean representations.



Mileage Outlier Handling (IQR Method)

Used the IQR technique to remove extreme mileage values that could distort downstream analysis.



Feature Engineering

Created meaningful features such as combined_fuel_economy, legroom, size_of_vehicle, and cleaned has_incidents to improve model readiness.



Final Dataset

Produced a fully cleaned, consistent dataset with no missing values and 19 key features ready for modeling.

Key Achievements So Far

✓ Clear Business Problem Defined

Understanding what affects used-car price and sale speed.

✓ Complete Data Understanding Performed

Explored structure, quality issues, missing values, outliers, and patterns.

✓ Full Data Preparation Completed

Clean dataset with engineered features ready for modeling.

Thank You

 For Your Attention 

Questions?

 We'd be happy to help

 contact@usedcarsdataset.com

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